

The Leibnizian Theory of Representativity of the Sign

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The aim of this paper is to distinguish, to characterize and to clarify the various models of representativity of the sign thought out by Leibniz in the context of his attempt to construct a Philosophical Universal Language. However, before doing so, we must try to locate our subject within Leibniz's linguistic thought and to point some preliminaries which we believe are necessary for a better understanding of our arguments.

1. It is in the *De arte combinatoria* (1660) that, for the first time, Leibniz defines the methodological regime that should govern the construction of the Universal Language in there thought as one of the applications of the "ars" (cf. GP 4.72-73).

Such a regime, which would essentially be reaffirmed in later formulations, supposes, as its first stage, the determination and exhaustive inventory of all the primitive ideas into which all our concepts can be resolved; second, the constitution of an alphabet of human thoughts through the attribution of signs to the primitive ideas; third, the regulated formation of the complex terms and their corresponding characters. On this view, the exhaustive inventory of our primitive ideas is the basis for the very possibility of constitution of a lexicon capable to adequately mirror the relations between ideas and to signify the totality of human knowledge.

It is well known that Leibniz has done various attempts to draw up that inventory — reducing it to the idea of God alone (cf. C 513), to the idea of God and Nothingness (cf. C 430), giving examples of numerous series (cf. GP 4.70-71, C 358-360, 514-515 or GP 2.542) or finding some pragmatic sol-

ution which would avoid the failure of the project, immediately on the level of its initial stage.¹

All these various attempts demonstrate the crucial importance Leibniz attributed to this first moment in which he locates the semantical foundation of the Universal Language to be constructed — a foundation which is here thought as the return or the opening of the symbolic system to the universe of intuition and, as such, to the world which is to be spoken.

It is clear that -as was inevitable- Leibniz never compiled the definitive, exhaustive list of primitive ideas, nor has he accepted none of his predecessor's proposals in this area, which he blamed to be arbitrary and deriving from analytical insufficiencies (cf. GP 3.216, GP 7. 19-20, 33-35. Cf. also Pombo 1987:79-81). However, this does not permit us to immediately conclude that it was as a result of such insuperable difficulties that Leibniz had opted for other models and developed other lines of research.²

This is not the place nor is it our aim to establish any causal relationship of this kind — which, moreover, we believe to be inadequate to the very style of Leibniz's thought. As a matter of fact, in the project for a Universal Language, as in other areas of his work, Leibniz frequently follows diverse paths and perspectives exploring, at the same time, a plurality of projects and strategies.

What interests us to stress is that there is another way of conceiving the semantical foundation of the Universal Language, a second alternative-strategy pursued in parallel throughout his life and which, furthermore, if well succeeded, could reinforce or even substitute the first way mentioned above.

We refer to the representativity of the sign that Leibniz wanted to conquer for the characteristic signs of the Universal Language, that is, his aim that (besides operativity or functional capacity of the signs within the formal system) signs should also be natural, representative or directly opened to the reality they are supposed to say. In both cases, that is to say, with this requirement for representativity as in the exigency of an inventory of primitive ideas, Leibniz aims to safeguard an extrinsic foundation of the symbolic system as a way of avoiding the complete formalism of a well made but empty language and to guarantee the prospective and heuristic role of the Philosophical Language.

However, with this requirement of the representativity of signs that foundation would now be achieved, no longer by the opening of the system to the ideal universe given by the intuition of the simple primitive ideas

(which are at the root of the complex concepts with which we usually think) but at an immediate, sensitive level, by the isomorphism between the characteristic signs (and their already composite characters) and the realities they signify; not then at a primal, intelligible level, but at a composite, sensible level, that of the motivation of the signs or of its representativity.

We can, therefore, infer that there are, in Leibniz, two strategies for the semantical foundation of the Universal Language — an intelligible and a sensitive one, we could say; and that it is on the level of the second strategy — that of the representativity of the sign — that, we believe, other important distinctions can be made. Their establishment is precisely what constitutes, the aim of this paper.

Two more points: 1) the critique, which has always signalized the importance of the Leibnizian requirement of the representativity of the characteristic signs, usually presents it as an homogeneous theory: a consequence of Leibniz's ideographical project — as in the case of Couturat (1901:88-89, 107-114) — or as the linguistic corollary of his theory of expression — as in the case of Knecht (1981:137 ff.) — just to refer two classics; 2) Leibniz never explicitly thematizes the differences between the several models which, we believe, he has thought out. On the contrary, he seems to be always speaking of the same, thus inviting the critique to the illusion of identity referred to above.

For our part, we believe, as it will be shown later, that it is possible, legitimate and even necessary to establish those differences. However, if we try to identify, to distinguish and to characterize different models within the leibnizian theory of the representativity of the sign (by reading, almost literally, we could say, his various, heterogeneous indications concerning this issue), it is not to stress the dispersive, non-systematic nature of Leibniz's thought but, on the contrary, so that we can recognize the powerful systematicity which always underlies the polymorphic nature of his work.

2. We propose indeed to distinguish *three principal models* in terms of which Leibniz intended to conceive the representativity of the sign.

The first model is the best known; the one to which, some critique, tends to reduce the all others). It appears for the first time in the *De arte* and it is still presented in the *Nouveaux Essais*. We refer to those famous passages in which, for instance, Leibniz writes:

"Et on pourroit introduire un Caractere Universel fort populaire et meilleur que le leur (he means, the Chinese), si on employoit de petites figures à la place des mots, qui representassent les choses visibles par leurs traits, et les invisibles par des visibles qui les accompagnent". (GP 5.379)

Representativity is here conceived in its strongest sense, that is, as the spacial and imagetive reproduction of the most pregnants traits of the signified reality. It is for that reason that we propose to term it as *figurative representativity*.

It may also be noted that this is a procedure which Leibniz believes to be applicable to all kinds of reality (visible and invisible...) and with which he also aims to guarantee even the figuration of syntactic relations when he suggests that the figures could be accompanied, as in Wilkins' system, by small, additional marks (cf. Couturat 1901:548-552). That is to say, we are here face to face with an extreme proposal of representativity which, obviously, would encounter insuperable difficulties in its realization.

This is something that, in other texts, Leibniz fully recognizes:

"<...> atque quae <res> pingi non possunt, qualia sunt intelligibilia, ea pinguntur tamen hieroglyphica quadem ratione sed eadem et philosophica (...). Quod fit, si non ut pictores, mystae aut Sinenses similitudines quasdam sectemur, *sed rei ipsius ideam sequamur*" [those realities which, like the intelligibles, cannot be figured, should nonetheless be represented by some hieroglyphic method (...). This can be done if we do not look for particular similarities as the painters, mystics and chinese do, but follow the idea of the thing itself]. (GM 5.216, our emphasis)

Clearly, Leibniz pursues here another model of representativity. What is now essential is not the figural representation of particular traits of sensory reality as it immediately appears to our perception, but the representation of the thing itself, that is, of the essential basis from which those same particularities can be deduced. That is why we propose to term this second model as *essentialist representativity*.³ Leaving aside the metaphysical problems underlying such a claim to a direct designation of the essence,⁴ one may even distinguish, in our opinion, two different orientations within the model in question.

The first, which we propose to call *genetic*, aims to represent essence on the basis of its status as causal principle.

We are referring here to the leibnizian project of constituting a Characteristic on the basis of a binary arithmetic, since he finds in the latter an admirable similarity between its mode of generation of numbers and the

relation of dependence of creatures with respect to God, their ultimate causal principle.⁵

In spite of the extremely vague and imprecise nature of Leibniz's indications concerning this possibility, we believe it is legitimate to suppose that, if it were possible to achieve, for the Characteristic, a kind of representativity similar to the one which, according to Leibniz, exists in binary arithmetic (in which the essence of each creature corresponds, as if it were, to its ordered place in the process of creation), then, it would be equally possible to deduce from each character the totality of properties of the creature it represents as a manifestation of its dependency on the first cause or absolute essence.

The second orientation, which we propose to term *analytic*, aims that the character should represent less the genesis than the composition of the idea signified. The essence is no longer thought as a generative principle but is now fundamentally conceived as the "Key" ("clavis") or unifying principle of a multiplicity of properties. We are referring to those texts in which Leibniz exposes the project of what he calls a "characteristica realis":

"Nomem tamen quod in hac lingua imponetur, *clavis* erit eorum omnium quae de auro humanitus, id est ratione atque ordine sciri possunt, cum ex eo etiam illud appariturum sit, quatenus experimenta de eo cum ratione institui dedeant" [The name which, in this language, should be given to gold, will be the key of everything that can be known about gold by humanity, that is, by reason and according to such an order that, from the examination of that name, it would be possible to deduce even those experiences which should be rationally achieved with it]. (GP 7.13, our emphasis)

The representativity of such a complex character, would therefore imply a process of analysis of the sign itself, which would identify in it, both the constitutive elements of the represented idea and the law of their combination. Leibniz argues that it is especially the analysis into prime factors that most closely corresponds to this aim⁶ given that, by representing the constitutive elements by undecomposable prime members and by taking their product as an analogical representation of the combination of the elements, it would be possible to discover the characteristic numbers of each of the constituent elements of that idea (cf. C 277).

There is, finally, a third model of representativity which aims to discover, neither the figuration of sensory particularities of the signified reality (as in the first model), nor the symbolization of the essential basis of those

particularities, whether as their causal principle, or as "Key" or principle of their unification (as in the second model, respectively in the genetic and analytic representativity), but rather the analogical and structural reproduction of the network of relations constituting the idea and its articulations. It is for this reason that we propose to term this model as *expressive representativity*.

It is clear that if one uses the term rigorously, all representativity is in Leibniz expressive, since expression is thought as the supreme instance of any type of relation. If, instead of considering representativity as an instance of the relation of expression, on the contrary, we view the latter as an instance of the former, this is because our aim is precisely to stress the multiple distinctions which this relation implies and which Leibniz subtly draws. Also, we aim to avoid the interpretative, critical tendency that stresses the expressive nature of the entire relation of representativity in Leibniz, thereby tending to blur all the differences which this relation implies.

Returning to the analysis of the expressive model, what must be noted, in the first place, is the fact that it is now geometry which provides the privileged model for the relation of expression. In fact, geometry is characterized by the immediate establishment of an isomorphism between the idea represented, as an ideal structure, and its schematic figuration; this isomorphism implies a search, not for similarities, but rather, as Leibniz puts it, for "un rapport constant et regle entre ci qui se peut dire de l'une et de l'autre" (GP 2. 112). In contrast to the reproductive model of figurative representativity, in which, as we have seen, absolute primacy is given to similarity (as the imitation of an original), in the case of expressive representativity that which expresses does not necessarily have to be similar ("simile") to the thing expressed, provided that some kind of analogy can be discerned in their form of being:

comme une Ellipse exprime un cercle vu de travers, en sort qu'à chaque point du cercle il en reponde un de l'Ellipse et vice-versa, suivant une certaine loy de rapport. (GP 1. 383).

That is, the relation of expression does not require similarity (although it does not exclude it either). It can even accept dissimilarity:

un même cercle peut être représenté par une ellipse, par une parabole, et par une hyperbole, et même par une autre cercle et par une ligne droite, et par un point. Rien ne paroît si different, ny si dissemblable, que ces figures; et cependant il y a un rapport exact de chaque point a chaque point. (GP 6.327; Cf. also GP 7. 263, C 15)

In a word, what the relation of expression absolutely demands is a structural and analogical parallelism.

Now, as early as in the *De arte*, Leibniz points towards such a model of expressive representativity, even if he then conceives it, in a purely *diagrammatic* form:

Commodum autem erit notas quam maxime fieri naturales, v.g. pro uno punctum, pro numeris puncta, pro relationibus Entis ad Ens lineas [It would be convenient that the signs would be as natural as possible, such as: for the one, one point; for the numbers, points; for the relations between one entity and another, lines]. (GP 4.73).

Each sign would then be emptied of any autonomous signification or any immediate reference to the real universe, and reduced to a simple pole of convergence of a multiplicity of relations, which would be represented by various types of lines according to their characteristics.

The same indication seems equally to be present in a brief passage of the fragment *Essais d'analyse grammaticale* (1683/4), where Leibniz considers to be licit to connect the components of the characters by different lines since, in this way, he says, it would be possible "to see" them, simultaneously, on the paper, that is, on the contrary to what happens with speech-sounds which disappear (cf. C 285). It should be noted that the recognition of the value of writing in contrast to speech presented here, it is not limited to the stressing of its advantages as a means of fixing and registering, or as a support for the failures of attention or reasoning. What Leibniz here emphasizes is the two-dimensionality of writing, and the consequent possibility of simultaneously grasping the multiple of relations which characters (and the ideas that they signify) establish among themselves. However, if the diagrammatic symbolism, to which the passages cited above tend, might have the merit of permitting the simultaneous apprehension of the totality of relations of the entities represented, it would shift the Universal Language in the direction of the pure formalism of a system of signs (lines, points, etc) devoid of any reference to the immediate universe of intuition and, therefore, irreducibly estranged from the world, thus contradicting the precise objectives at which Leibniz aims with the very representativity of the sign.⁷

There is, we believe, another subtly different tendency of thinking this expressive representativity in the celebrated *Dialogus* of August, 1677, when Leibniz writes:

Nam etsi characteres sint arbitrarii, eorum tamen *usus et connexio* habet quiddam quod non est arbitrium, scilicet *proportionem* quandam inter characteres et res, et diversorum characterum, easdem res exprimentium relationes inter se [Even if characters are arbitrary, their use and connection have however something which is not arbitrary, that is, a certain proportion between characters and things and, at the same time, between the various characters which express the relations of things among themselves]. (GP 7.192, our emphasis).

Leibniz is now forced, as if it were, to displace the ideal of representativity, from the character (without, however, completely abandoning it), on to the plane of syntactic relations. The meaning of each character comes now to depend either on its integration in a formal structure — its “*usus et conexio*” — or on the correspondence or “*proportio*” between this structure and the structure of the real which it aims to represent, that is to say: representativity would now be played out fundamentally on the level of syntactic structures.

The critique has often commented upon this important text, frequently seeing in it the very model of representativity of the sign thought out by Leibniz. However, according to our proposal, we would be dealing here with a slightly different orientation or subtype within the expressive model. We could term it as *structural representativity in the strict sense*, since here, it is the very form of relations between characters that is seen as analogical with things in the form of their relations.⁸

Of course, the complex problem of the ontological status of relation, clearly remains to be solved. Are relations real entities or “mere mentale” (cf. GP 2.486)? In the later case, how could a “mere mentale” entity constitute the anchorage point of the multiple relations established within the proposition? And what, then, would be the ontological correlatum of the syntactic relations?

On the other hand, if Leibniz can conceive relation as a “mere mentale”, it is because he reduces it to a series of modifications of singular beings as substantial nuclei of their own predicates. So, we could ask, would it not also be possible, on the level of language, to similarly view signs as poles of irradiation of syntactic relations, reducing the latter, in the same way, to different types of modification of the former? But, in this case, we would again be face to face with another subtle difference: the expressive representativity of such signs would again come into being, not between structures (syntactic and natural), but between elements: between the singular individual (containing relations as its predicates) and the

character as a sign which, through its own flexion, would "exhibit" the multiple relations of the thing.

And, indeed, it is Leibniz who says in the fragment *Characteristica Geometrica*, (10 August 1679):

<...> et si quando *exhibeant* omnes rerum relationes inter se, (...) nihil erit in re, quod non per characteres deprehendi possit [if the characters could exhibit all the relations of the things, (...) then in the things it would not be anything which could not be deduced from the characters]. (GM 5.141, our emphasis).

It is now again each character, and not only the relations between characters, that is to express the relations which individual entities contain as their predicates. It is for that reason that, further on in the same fragment, Leibniz defines characters as "<...> quibus aliarum rerum inter se relationes exprimuntur". (GM 5.141).

Then, it is possible to distinguish, within the expressive model, the three tendencies which we have just expounded. In the first tendency, the structure is purely formal; in the second, the structure becomes a source of meaning because, in itself, it imitates the real conceived as a structure; in the third, the structure is absorbed by the sign, and it is through the sign that it becomes expressive.

If in the diagrammatic tendency the semantic level tends to be entirely reduced to the syntactic and, in the strictly structural tendency, priority is clearly given to the analogy between the formal structures of language and the structures of the real, the third and last tendency recovers (from the figurative and essentialist models) the requirement of the *semantics of the sign itself* with which Leibniz recovers a long tradition coming from the speculations on the Adamic Language and passing through all the Middle Ages and Modern Times (cf. Dubois 1970; Kayser 1972 and Pombo 1987: 33-52).

In the diagrammatic model of expressiveness, the significative elements are reduced to empty poles of convergence (points) of an abstract multiplicity of relations (lines), thus sliding the project towards a radical formalism which Leibniz cannot accept; in the structural model of expressiveness, it is syntax itself, through its analogy with the supposed structure of facts, that tends to become semantic; while, the third and last model of expressiveness, which may perhaps be termed *monadological* (given that the character absorbs the entire expressive function into itself, becoming the stable point of semantic organization) recovers the requirement of a space that should

be minimally filled, symbolically differentiated and, as such, indicative of the signified reality.

We are left, then, again with that enigmatic indicative power of the sign which Leibniz aims to safeguard at all costs, and which the various models of representativity that we have discussed are intended to elucidate.

3. Two last points: 1) None of the precedent models of representativity of the sign is without problems. On the contrary, none of them provide satisfactory solutions for the problems that they intended to solve; 2) It remains to be said that those problems continue to quarel us today almost as much as in Leibniz's time. That is why, we believe, the interest in discussing this subject and in trying to discover such type of differences is not only *internal* but also *external* to the exegesis of Leibniz's linguistic thought.

Internal because the interpretative hipotesis we have been proposing may permit to recognize one possible unifying pole for the three great and apparently contradictory Leibniz's linguistic projects: a) the study (and perfection) of natural languages (german in especial), in which Leibniz looks for that representativity, discovering it both at the level of the etimologies and on the present motivation of its vocabulary; b) the search for the constitution of a Rational Grammar on the basis of the deep universal structure which is expected to underly the grammatical particularities of the various national languages; and c) the construction of a Universal Language endowed with a similar or even higher representative nature, and in which the representativity of its signs would somehow repeat (and even perfect) something that had already been achieved in natural languages.⁹

External in the sense that this leibnizian theory of the representativity of the sign (and its various models) may become, in near future, extremely helpful for us to think the question of the representative /versus arbitrary nature of the human languages. A question apparently closed, early in our century with the Saussurian postulate of the arbitrariness of the sign, but which, certainly, will be soon reopened. A question too whose history would be interesting, and extremely useful, to remake in the field of the History of Language Sciences.

When one looks to the Leibnizian theory of the representativity of the sign, it is somehow necessary to forget what we know today, or what we suppose to know, not because we have attained today a thruth that Leibniz could not have reached in his time but, inversely, because Leibniz thought thoughts and pointed to hypothesis to which our present rationality is

closed. Indeed, Leibniz is, neither a predecessor of some of the thesis of the language sciences of our days, nor is he someone who had not yet acceded to our present scientific understanding of the human language. The fact is that our present sciences of language are just one *means*, among others, to look for the elucidation of the intimate nature of human language. A means that, precisely because it looked for becoming a science, had to circumscribe its own field under study, to delimitate its own object of investigation and then, to leave out hypotheses, subjects and questions which probably were undeservedly forgotten.

But we cannot loose the global sense of the problem, which, certainly, Leibniz has much more deeply thought than we do.

NOTES

1. See, for instance, the distinction established by Leibniz between the "termes absolutement primitifs" and the "termes primitifs a nostre egard" (C 176) from which he had admitted one could start out (Cf. also C 220-221).
2. On the contrary, he argued against Descartes on this precise matter. Indeed, unlike Descartes for whom "l'invention de cette langue depend de la vraie philosophie" (AT 1.81), Leibniz, although recognizing the difficulty pointed out by Descartes (Cf. also *letter to Burnet, August, 24th, 1697*, GP3.216), defends: "<...> quoyque cette langue depende de la vraye philosophie, elle ne depend pas de sa perfection. C'est a dire, cette langue peut estre établie, quoyque la philosophie ne soit pas parfaite" (C 28). Cf. also Pombo (1987:95-100).
3. If rigorously conceived, all representativity is essentialist since, according to Leibniz, the sign represents, not a particular entity or concept but an essence or, as Leibniz says: "la possibilite de ce qu'on propose". (GP 5.272).
4. It would be necessary to choose one of the multiple perspectives through which the essence can be expressed and besides, to determine it in real and not just in nominal terms.
5. As Leibniz says in the *Remarques sur Weigel*: "Car, d'après la dyadique, tous les nombres sont exprimés par les seules caractères O et I, par l'unite et O; remarquable analogie de la création des choses sorties de Dieu et du néant" (FNL 167). Cf. also C 430 and GM 7.223-2127. Cf. too Dascal (1978:218-221).
6. Leibniz also considers chemical symbols and Egyptian and Chinese hieroglyphics as examples of the "characteristica realis". Cf. GP 7.12.
7. The similarities between Frege's ideography and the Leibnizian project of a Characteristic Language would be especially established, in our view, with this diagrammatic form, given that Frege rejects the close articulation established by Leibniz between the Universal Language and the Encyclopedia, leaving to the sciences the labour of defining their

own concepts. For the relationship between Frege's ideographic project and the leibnizian project of a *characteristica universalis*, Cf. Kluge (1979:21-29) and Imbert (1979:621-665).

8. Perhaps one could defend the proximity between this Leibnizian theory and that which, two hundred years later, the young Wittgenstein would present in the *Tractatus* — the power of a picture or logical proposition derives from the fact that it shares the same "form of representation" with reality. Cf. Kneale (1966:204-215) who studies that proximity. Cf. also Kretzmann (1964:382).
9. That is precisely the conclusion we reach in Pombo (1987), where this interpretative line of research is pursued in detail, namely through the comparative study of the three great leibnizian linguistic projects: the perfection of a natural language, the a posteriori construction of a Rational Grammar and the a priori construction of a new Philosophical Language.

REFERENCES

- AT *Oeuvres de Descartes*. Ed. by Charles Adam and Paul Tannery. Paris: Vrin, 1964-76.
- C *Opuscules et fragments inédits de Leibniz. Extraits des manuscrits de la Bibliothèque royale de Hannover par Louis Couturat*. Hildesheim: Olms, 1961.
- FNL *Nouvelles lettres et opuscules inédits de Leibniz, précédés d'une introduction par Louis Alexandre Foucher de Careil*. Hildesheim/New York: Olms, 1971.
- GM *G.W. Leibniz. Mathematische Schriften*, hrsg. v. Carl Immanuel Gerhardt, vols. 1-7. Hildesheim: Olms, 1962.
- GP *Die philosophischen Schriften von Gottfried Wilhelm Leibniz*, hrsg. v. Carl Immanuel Gerhardt, vols. 1-7. Hildesheim: Olms, 1960.
- Couturat, Louis. 1901. *La logique de Leibniz d'après des documents inédits*. Paris: Alcan.
- Dascal, Marcelo. 1978. *La sémiologie de Leibniz*. Paris: Aubier-Montaigne.
- Dubois, Claude-Gibert. 1970. *Mythe et langage au seizième siècle*. Bordeaux: Ducros.
- Imbert, Claude. 1979. "Le project idéographique". *Revue Internationale de Philosophie* 130. 620-665.
- Kayser, Wolfgang. 1972. "La doctrine du langage naturel chez Jacob Boehme et ses sources". *Poétique* 11. 337-366.

- Kluge, Eike-Henner W. 1979. "Frege, Leibniz and the notion of an ideal language". *La Filosofia en America. IX Congreso Interamericano de Filosofia*, II. 21-29. Caracas.
- Kneale, William Calvert. 1966. "Leibniz and the Picture Theory of Language", *Revue Internationale de Philosophie*. 76/77. 204-215.
- Knecht, Herbart. 1981. *La logique chez Leibniz. Essai sur le rationalisme barroque*. Lausanne: L'age d'homme.
- Kretzmann, Norman. 1964. "History of Semantics". *The Encyclopedia of Philosophy*, vol.7. 358-406. New York: Macmillan Free Press/London: Collier-Macmillan.
- Pombo, Olga. 1987. *Leibniz and the problem of a Universal Language*. Münster: Nodus Publikationen.

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55. HALL, Robert A., Jr.: *A Life for Language. A biographical memoir of Leonard Bloomfield*. Amsterdam/Philadelphia, 1990.
56. VERSTEEGH, Kees and Michael G. CARTER (eds): *Studies in the History of Arabic Grammar II*. Amsterdam/Philadelphia, 1990.
57. STARNES, de Witt T. and Gertrude E. NOYES: *The English Dictionary from Cawdrey to Johnson 1604-1755. Reprint of the 1946 edition, updated and with an introduction by Gabriele Stein*. Amsterdam/Philadelphia, 1990. n.y.p.